A. **Floor Function.** Modular arithmetic isolates the remainder of a division.

The floor function will isolate the quotient.

The **floor function** is defined for real numbers x by

$\lfloor x \rfloor =$

So for example:

$\lfloor 9.213 \rfloor =$

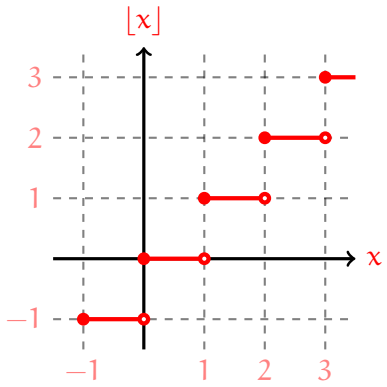
$\lfloor -5.0001 \rfloor =$

$\lfloor 11 \rfloor =$

We quickly conclude

$\lfloor x \rfloor = x$ if and only if x is:

Here is a graph of the floor function.



The part that we "lose" when we take a floor is

The **fractional** part of a real number x is

$\{x\} =$

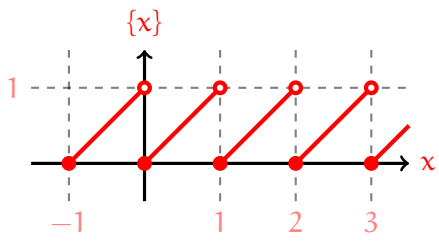
So for example:

$\{9.213\} =$

$\{-5.0001\} =$

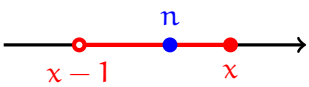
$\{11\} =$

Here is a graph of the fractional part.



B. **Floor Function Properties.** Here is a characterization of the floor function.

Theorem. Let x be a real number and n an integer so $x - 1 < n \leqslant x$.



Then $\lfloor x \rfloor =$

Proof. Evidently n is:

The next largest integer is $n + 1$, but since $n > x - 1$ it follows that

$n + 1$

Hence $n + 1$ not less or equal x.

Hence the greatest integer less or equal x is:

□

And here is how we can relate floors to divisions.

Theorem. Let a and $b > 0$ be integers and execute $a \div b$ to obtain

$a = qb + r$ where q, r are integers and $0 \leqslant r < b$

Then: $q =$

$r =$

Proof. First divide $a = qb + r$ by b, then solve for q, to obtain

Since further q is an integer, by the theorem at the top of the page

$\left\lfloor \frac{a}{b} \right\rfloor =$

We next calculate $\left\{ \frac{a}{b} \right\} =$

□

Let's exhibit this for $166 \div 20 = 8.3$.

C. **Floor and Factorials.** The floor helps find prime factorizations of factorials.

Theorem. Let n be an integer and p be a prime.

Then the exponent of the largest power of p that divides $n!$ is

Before we prove it, let's do a quick check.

According to this theorem, the largest power of 2 that divides $10!$ equals:

$$\sum_{k=1}^{\infty}$$

which is confirmed by the prime factorization of $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$.

We could have used the same theorem to find the exponents of 3, 5, and 7 that we have provided.

Proof. Because p is prime, and $n! = 1 \cdot 2 \cdots n$, we can answer this by

The number of multiples of p in the list

$1, 2, \ldots, n$

is given by:

and that many p's are contributed as:

The number of multiples of p^2 in the list

$1, 2, \ldots, n$

is given by:

and that many additional p's are contributed as:

Continuing this process with p^2, p^3, and so on, the total $\#$ of factors of p in $n!$ is

□

Example 1. Find the number of zeroes in which $53!$ terminates.

Solution. We are looking for

But our theorem only applies for prime powers.

If we can find the maximal power of k of 2 and l of 5 that divide $50!$, then

$2^k 5^l \mid 53!$

and thus the maximal power of 10 that divides $53!$ is:

Next, according to our theorem, as factors of factorials, smaller primes

Thus, the maximal exponent of 10 matches the maximal exponent of:

That exponent is